



- ✓ 1. (a) The Maxwell-Boltzmann distribution may not be quite valid! Calculate the fraction of N_2 molecules having a speed in excess of the speed of light. (3 Marks)
(b) Justify your answer with a solid explanation (2 Marks)

[Hint: $f(v)dv = 4\pi(m/2\pi k_B T)^{3/2} v^2 \exp(-mv^2/2k_B T) dv$

Mass of N_2 : 28 amu; $T = 300K$]

- ✓ 2. For 1.0 mol of $CH_4(g)$ at 0 degrees Celsius and 1 atm, find the number of molecules whose speed lies between the range of 90.000 m/s to 90.002 m/s. (5 Marks)
[Hint: $dN_v/N = f(v)dv$; mass of $CH_4 = 16 g/mol$]

- ✓ 3. Write down the units of the rate constant: (5 Marks)
(a) zero-order
(b) half-order
(c) first-order
(d) 3/2-order
(e) second-order

- ✓ 4. *p*-nitrophenylglutamate undergoes hydrolysis at 298K. The reaction is followed spectrophotometrically at 400 nm, with a cuvette having a path length of 1 cm. If the A_{400} of the reaction is 0.634 at pH 7.14, calculate the first-order rate constant. (3 Marks; graph 2 marks)

t(sec)	A_t	t(sec)	A_t
0	0.0	120	0.402
40	0.265	180	0.451
80	0.342	α	0.634

[Hint: use a proper graph paper to estimate the rate constant]

Important Formulae and universal constants:

1. $h = 6.626 \times 10^{-34} Js$

2. $k_B = 1.38 \times 10^{-23} J/K$

3. $c = 2.998 \times 10^8 m/s$

4. $N_A = 6.023 \times 10^{23} /mol$